

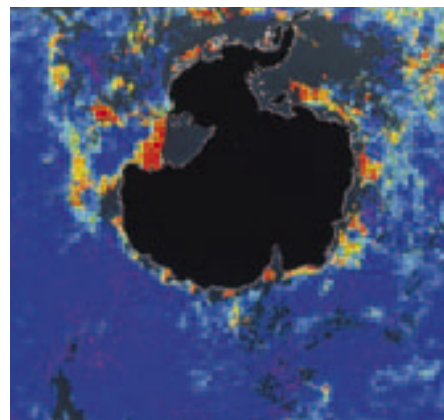


LIFE IN ICY OCEANS

The shores of polar oceans are mostly barren, bare rock, because the sea ice that forms in winter destroys most shore life. Around Antarctica, seals and penguins follow the edge of the sea ice as it expands away from the coast, so they can hunt in open water. This takes them far out into the Southern Ocean. In the north, ice covers most of the Arctic Ocean in winter, so animals congregate around the few remaining patches of open water. But when the ice melts in summer, the cold waters teem with life as plankton multiplies in the almost perpetual daylight.

▲ BARREN BEACHES

Frozen solid in winter, and scoured by moving ice as the tide rises and falls, many of the shores of polar oceans are bare wastes of rock and gravel. But this does not concern animals that feed in the sea, such as these Arctic walrus, which use them to rest and warm up after hunting for shellfish in the cold water. Their pinkish brown skins contrast with the gray of the few animals that are still cold.



RICH SEAS ►

In shallow polar seas, storms stir up nutrients from the seabed and mix them with the surface water. This enables plankton to multiply rapidly and create a wealth of food when the pack ice breaks up, as shown in red and yellow in this satellite image of Antarctica in summer.



▲ SWARMING KRILL

Around Antarctica, the multiplying phytoplankton is harvested by vast swarms of shrimplike krill that can cover huge areas of ocean. They are the main prey of most of the animals that live on and around Antarctica. These include huge whales that use the sievelike bristles fringing their enormous mouths to strain the krill from the water. Krill also support many penguins, seabirds, and seals.

LEADS AND POLYNAS



In the winter and spring, the few stretches of the Arctic Ocean that do not freeze over act as magnets for marine mammals like these narwhals. They find all their food in the water, but they must be able to surface to breathe, and this is much more difficult where the ocean surface is frozen. Broad polynas (large areas of open water surrounded by sea ice) and narrower leads (like the one seen here) also attract fish-eating seabirds in spring. Most polynas open up in spring as the pack ice starts melting, but some—such as Northwater between Canada and Greenland—stay open throughout the winter.